

Marcel Torne Villasevil

PhD Student, Stanford Artificial Intelligence Laboratory (SAIL) | IRIS Lab • Advisor: Prof. Chelsea Finn

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* indicates equal contribution / equal supervision.

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EDUCATION

Stanford University

Sept. '24 – Jun. '29 (exp.)

PhD in Computer Science, Stanford, CA

Stanford Artificial Intelligence Laboratory (SAIL), IRIS Lab.

Advisor: Prof. Chelsea Finn.

Harvard University

Aug. '21 – May '23

M.Eng. in Computational Science and Engineering, Cambridge, MA

School of Engineering and Applied Sciences (SEAS). GPA: 4.00/4.00.

Relevant courses: Statistical Decision Making (Prof. Susan Murphy); Stochastic Methods for Data Analysis, Inference and Optimization (Dr. Weiwei Pan).

Massachusetts Institute of Technology

Aug. '21 – May '23

Master's thesis and cross-registered student, Cambridge, MA

Computer Science and Artificial Intelligence Laboratory (CSAIL). GPA: 4.00/4.00. Master's thesis with Prof. Pulkit Agrawal and Prof. Abhishek Gupta at the Improbable AI Lab.

Relevant courses: Robotic Manipulation (Prof. Russ Tedrake); Sensorimotor Learning (Prof. Pulkit Agrawal); Advances in Computer Vision (Prof. Philip Isola, Prof. Bill Freeman); Natural Language Processing (Prof. Jacob Andreas, Prof. Yoon Kim).

École Polytechnique Fédérale de Lausanne

Sept. '18 – Jul. '21

B.Sc. in Computer Science, Lausanne, Switzerland

Grade: 5.62/6 (top 3%).

International Baccalaureate (IB Diploma)

Sept. '16 – Jun. '18

IES Gabriel Ferrater i Soler, Reus, Spain

Grade: 42/45 — highest score in the school's history.

CONFERENCE PUBLICATIONS

Freeform Preference Learning for Robotic Manipulation

Marcel Torne Villasevil*, Anubha Mahajan*, Abhijnya Bhat*, Chelsea Finn

Conference on Robot Learning (CoRL), 2026

[website](#) | [arXiv:2606.32027](#)

MEM: Multi-Scale Embodied Memory for Vision-Language-Action Models

Marcel Torne Villasevil*, Karl Pertsch*, Homer Walke, Kyle Vedder, Suraj Nair, Brian Ichter, Allen Ren, Haohuan Wang, Jiaming Tang, Kyle Stachowicz, Karan Dhabalia, Michael Equi, Quan Vuong, Jost Tobias Springenberg, Sergey Levine, Chelsea Finn, Danny Driess

Conference on Robot Learning (CoRL), 2026

[website](#) | [arXiv:2603.03596](#)

$\pi_{0,7}$: a Steerable Model with Emergent Capabilities

Physical Intelligence (incl. Marcel Torne Villasevil), et al.

Conference on Robot Learning (CoRL), 2026

[blog](#) | [paper](#)

$\pi_{0,6}^*$: a VLA that Learns from Experience

Physical Intelligence (incl. Marcel Torne Villasevil), et al.

Robotics: Science and Systems (RSS), 2026

[blog](#) | [paper](#)

Learning Long-Context Diffusion Policies via Past-Token Prediction

Marcel Torne Villasevil*, Andy Tang*, Yuejiang Liu*, Chelsea Finn

Conference on Robot Learning (CoRL), 2025

[Best Paper Award](#), Robot Representations Workshop, RSS 2025 | [website](#) | [arXiv:2505.09561](#)

Robot Learning with Super-Linear Scaling

Marcel Torne Villasevil*, Arhan Jain*, Jiayi Yuan*, Vidaaranya Macha*, Lars Ankile, Anthony Simeonov, Pulkit Agrawal, Abhishek Gupta

Robotics: Science and Systems (RSS), 2025

[website](#) | [arXiv:2412.01770](#)

From Imitation to Refinement: Residual RL for Precise Visual Assembly

Lars Ankile, Anthony Simeonov, Idan Shenfeld, Marcel Torne Villasevil, Pulkit Agrawal

IEEE Int. Conf. on Robotics and Automation (ICRA), 2025

[website](#) | [arXiv:2407.16677](#)

Reconciling Reality through Simulation: A Real-to-Sim-to-Real Approach for Robust Manipulation

Marcel Torne Villasevil, Anthony Simeonov, Zechu Li, April Chan, Tao Chen, Abhishek Gupta*, Pulkit Agrawal*

Robotics: Science and Systems (RSS), 2024

[website](#) | [arXiv:2403.03949](#) | [Media: TechCrunch, MIT CSAIL](#)

Breadcrumbs to the Goal: Goal-Conditioned Exploration from Human-in-the-Loop Feedback

Marcel Torne Villasevil, Max Balsells i Pamies, Zihan Wang, Samedh Desai, Tao Chen, Pulkit Agrawal, Abhishek Gupta

Advances in Neural Information Processing Systems (NeurIPS), 2023

[website](#) | [arXiv:2307.11049](#) | *Media: MIT News spotlight*

Autonomous Robotic Reinforcement Learning with Asynchronous Human Feedback

Max Balsells i Pamies*, **Marcel Torne Villasevil***, Zihan Wang*, Samedh Desai, Pulkit Agrawal, Abhishek Gupta

Conference on Robot Learning (CoRL), 2023

[website](#) | *Media: TechXplore*

PREPRINTS

PolaRiS: Scalable Real-to-Sim Evaluations for Generalist Robot Policies

Arhan Jain, Mingtong Zhang, Kanav Arora, William Chen, **Marcel Torne Villasevil**, Muhammad Zubair Irshad, Sergey Zakharov, Yue Wang, Sergey Levine, Chelsea Finn, Wei-Chiu Ma, Dhruv Shah, Abhishek Gupta, Karl Pertsch

arXiv preprint, 2025

[website](#) | [arXiv:2512.16881](#)

RoboArena: Distributed Real-World Evaluation of Generalist Robot Policies

Pranav Atreya*, Karl Pertsch*, Tony Lee*, Moo Jin Kim, Arhan Jain, **Marcel Torne Villasevil**, Abhishek Gupta, Chelsea Finn, Sergey Levine, et al.

arXiv preprint, 2025

[website](#) | [arXiv:2506.18123](#)

RESEARCH AND WORK EXPERIENCE

Stanford University — SAIL, IRIS Lab

Sept. '24 – Present

Graduate Research Assistant, advised by Prof. Chelsea Finn — Stanford, CA

- Memory architectures for vision-language-action models and long-context robot policies.
- Learning dense reward signals from rich, freeform human feedback for long-horizon manipulation.

Physical Intelligence

Jun. '25 – Mar. '26

Research Intern — San Francisco, CA

- Introduced memory into the π family of policies; core contributor to MEM and $\pi_{0.7}$.

MIT CSAIL — Improbable AI Lab

Feb. '22 – Aug. '24

Graduate Research Assistant, advised by Prof. Pulkit Agrawal and Prof. Abhishek Gupta — Cambridge, MA

- Leveraged crowdsourced non-expert human feedback from annotators worldwide to guide exploration.
- Developed a new algorithm for finetuning robots in the real world from human feedback.
- Used real-to-sim transfer of real scenes to robustify policies through simulation.

Microsoft

Jun. '22 – Aug. '22

Data Scientist Intern — Bellevue, WA

- Microsoft Bing: built the first production implementation of Microsoft Research's Extreme Classification algorithms in the Bing image ranker, ranking billions of images against text queries.

Amazon

May '20 – Sept. '20

Software Engineering Intern — Barcelona, Spain

- Amazon Search, Science and AI: built an AngularJS browser plugin to help ML engineers debug the Amazon ranking algorithm; used 100+ times per day by 20+ engineers.

EPFL — MLO and iGH Labs

Feb. '21 – Jul. '21

Bachelor's thesis, supervised by Prof. Martin Jaggi and Prof. Mary-Anne Hartley

- Started DISCO, an open-source platform for privacy-preserving decentralized machine learning, demonstrated on medical image classification.

EPFL — Formula Student Racing Team

Feb. '21 – Jul. '21

Autonomous driving team — independent research

- Trained a control and planning policy for an autonomous vehicle using deep RL and sim-to-real transfer.

TEACHING EXPERIENCE

Stanford University — CS224R, Deep Reinforcement Learning

Mar. '26 – Jun. '26

Head Teaching Assistant

Led the course staff for one of Stanford's largest graduate classes (~800 students); owned assignments, grading and the final project pipeline.

Stanford University — CS224R, Deep Reinforcement Learning

Mar. '25 – Jun. '25

Teaching Assistant

Delivered a tutorial lecture and ran weekly office hours.

EPFL — Computer Architecture

Sept. '20 – Jan. '21

Teaching Assistant

Led a team of a dozen graduate students and owned the major course project: implementing Conway's Game of Life in assembly.

EPFL — General Physics I: Mechanics

Sept. '19 – Dec. '19

Teaching Assistant

Led exercise sections for groups of 20–40 students and assisted with weekly problem sets.

SELECTED HONORS AND AWARDS

Best Paper Award — Robot Representations Workshop, RSS

Jun. '25

For Learning Long-Context Diffusion Policies via Past-Token Prediction.

Harvard Real Colegio Complutense Scholarship

Aug. '21 – May '23

Merit-based scholarship for Spanish students at Harvard; covered ~70% of the Master's tuition.

Harvard IACS Master's Thesis Proposal Award

Aug. '21 – May '23

Awarded to a small, selected number of students in the master's program for an outstanding thesis proposal; covered ~30% of second-year tuition.

Nova 111 Student List, Spain

May '22

Selected as one of the top 11 Computer Science students in Spain aged 18–25.

Excellence in the University Entrance Exam

Jul. '18

Top 100 of 33,000 students; award given by the Catalan president.

Bronze Medal, Spanish National Physics Olympiad

Jun. '18

City of Reus High School Academic Excellence Award

Sept. '16

Given by the mayor to the top student at each high school.

INVITED TALKS

Y Combinator Paper Club

Jul. '26

Presented our work on memory for VLA models. [Video](#).

ExWBC Workshop, RSS 2026 — Sydney, Australia

Jul. '26

Invited speaker.

IROS 2025 Workshop

Oct. '25

Invited 15-minute talk: *Real-to-Sim-to-Real Visuomotor Robot Learning*.

CoRL 2024 — Spotlight talk

Nov. '24

Google Developer Student Club, Lausanne

Nov. '22

Talk on my Microsoft internship and the US job-search process.

SERVICE AND REVIEWING

Reviewer: ICRA 2025, IROS 2025, RSS 2024, ICLR 2025 Workshop on World Models and Robot Learning, RSS 2025 Workshop on Dexterous Manipulation.

MIT Robotics Seminar — Student Volunteer and Co-host

Fall '23

- Hosted and toured invited faculty around MIT and organized their day (student roundtables, lunches, faculty meetings).
- Hosted Prof. Chelsea Finn (Stanford), Prof. Nima Fazeli (Michigan), Prof. Cynthia Yu (Penn), Prof. Yu Sun (Toronto).

MENTORSHIP AND OUTREACH

Research mentorship

'23 – Present

Supervise students across multiple research projects, meeting several times per week on both technical challenges and higher-level research guidance.

- Jerry Mao, MIT undergraduate — LLMs for robotics.
- Carrie Yuan, UW Master's — finetuning pretrained models.
- Samedh Desai, UW undergraduate — human-in-the-loop learning.

Project Moonshot — academic mentorship

'18 – Present

Mentor students from Spain through the process of applying to top programs in Europe and the US.

Volunteering for inclusion

2012 – '18

- *Nastic Genuine* (2018): helped organize a Spanish national football tournament for people with Down syndrome.
- *Regatta Carla for RETT syndrome* (2018): social media and promotion for a fundraising sailing regatta.
- *La Marató de TV3* (2012–'18): performed baroque guitar and voice to raise money for medical research.

OPEN-SOURCE PROJECTS

DISCO: Distributed Collaborative Machine Learning

'21 – Present

MLO and iGH Labs, EPFL — supervised by Prof. Martin Jaggi

Co-founded with Paul Mansat as my bachelor's thesis. An open platform for federated and decentralized model training, now used by graduate students at the MLO lab. github.com/epfml/disco (190+ stars) | [live demo](#).

FOUNDING EXPERIENCE

Quiz.Fit

Jun. '19 – Jan. '21

Co-founder and lead developer — Lausanne, Switzerland

Built an iOS app in Swift that improves users' health through gamification of healthy habits.

MEDIA COVERAGE

TechCrunch — *Researchers are training home robots in simulation based on iPhone scans*, Jul. 24, 2024.

MIT News (spotlight) — *New method uses crowdsourced feedback to help train robots*, Nov. 27, 2023.

MIT CSAIL (spotlight video) — *Precision Home Robotics with Real-to-Sim-to-Real*, Jul. 18, 2024.

University of Washington News — *Using photos or videos, these AI systems can conjure simulations that train robots*, Aug. 7, 2024.

TechXplore — *An approach that allows robots to learn in changing environments from human feedback and exploration*, Nov. 28, 2023.

Diari de Tarragona — *A mentor from Salou to help students get into Harvard and Stanford*, Aug. 2, 2024.

Additional mentions by **Franka Robotics** and **Six Degrees of Robotics**.

SKILLS

Languages: Catalan (native), Spanish (native), French (native), English (fluent), Portuguese (beginner), German (beginner).

Programming: Python, JavaScript, C, C++, Scala, Spark, Kotlin, SQL, HTML, CUDA, VHDL, MIPS assembly.

ML libraries: PyTorch, JAX, TensorFlow, PyTorch Lightning, Weights & Biases.

Robotics simulators: Isaac Sim, MuJoCo, PyBullet, Drake.